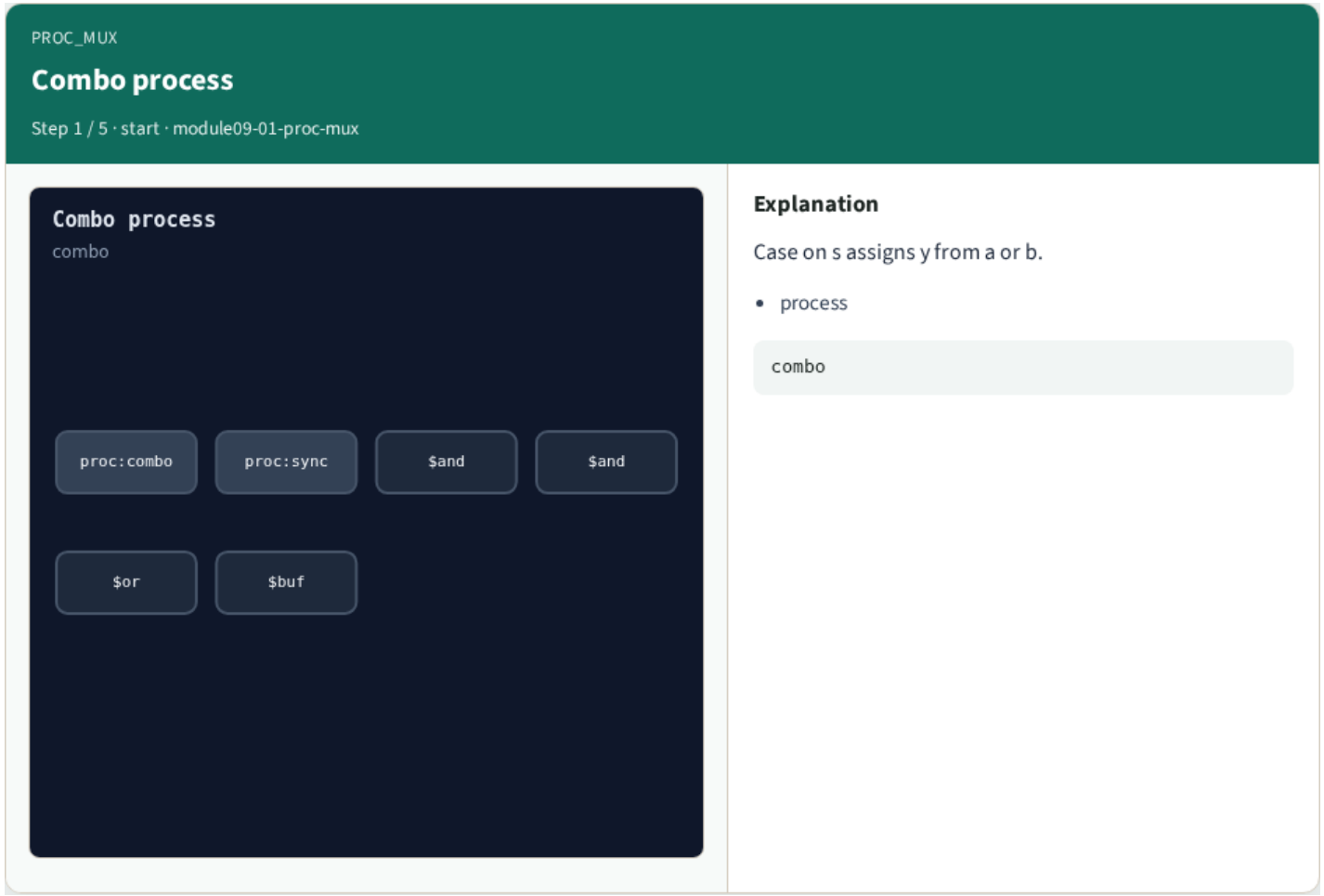
Why proc mux

Combo case and if processes become mux cells before registers appear

The idea

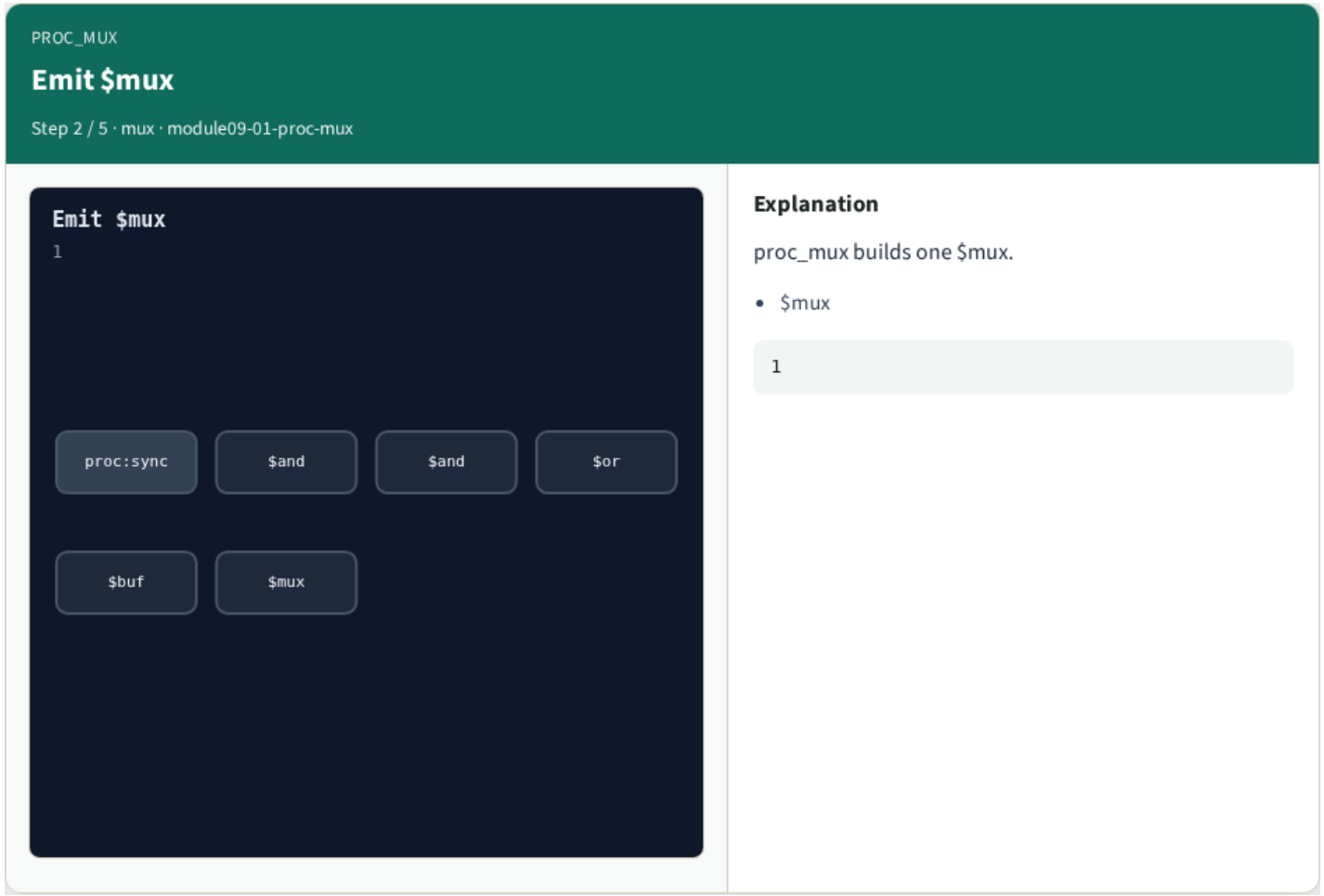
- GOLDEN: mux_count=1; type is dollar mux

Combo process



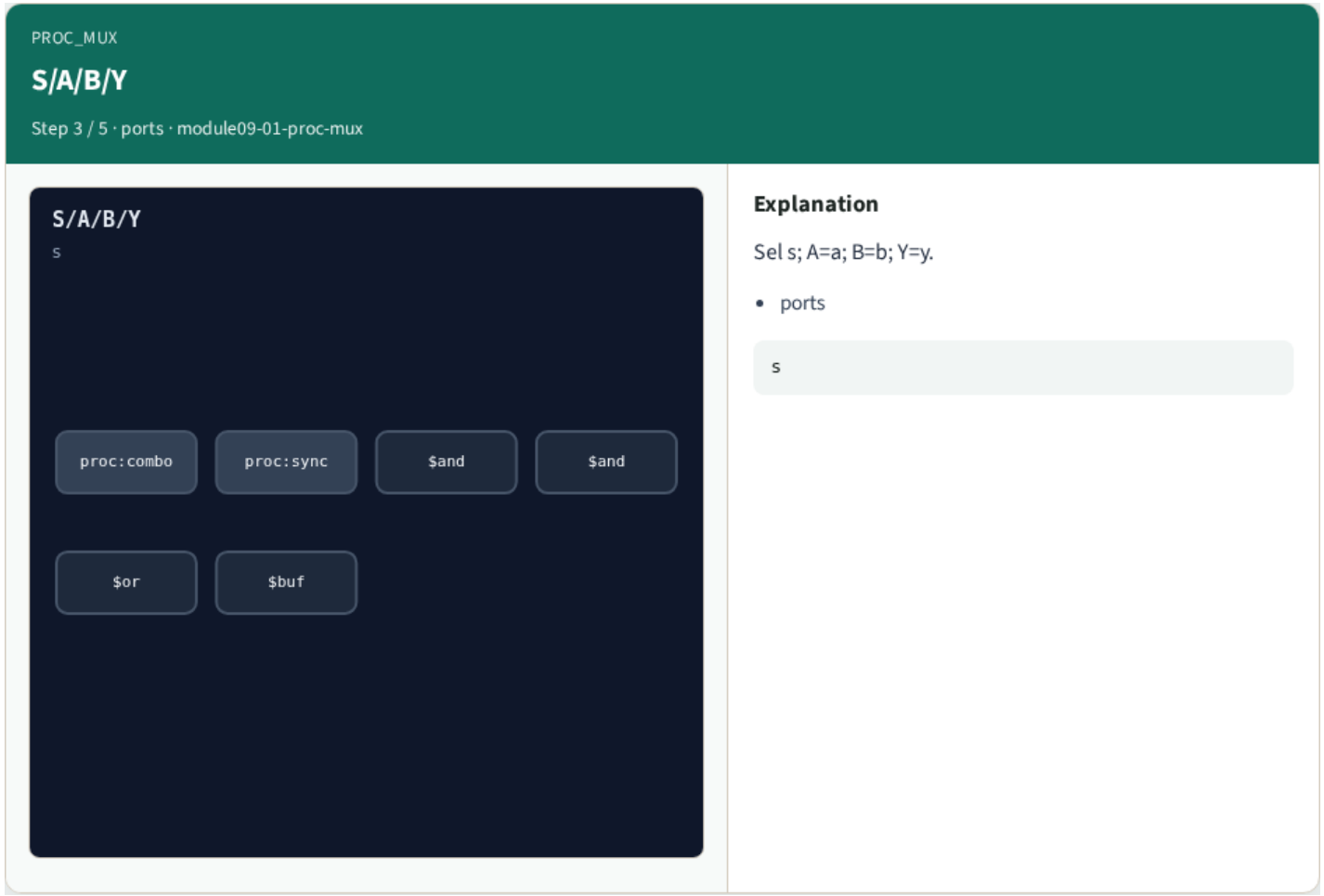
Combo process

Emit $mux



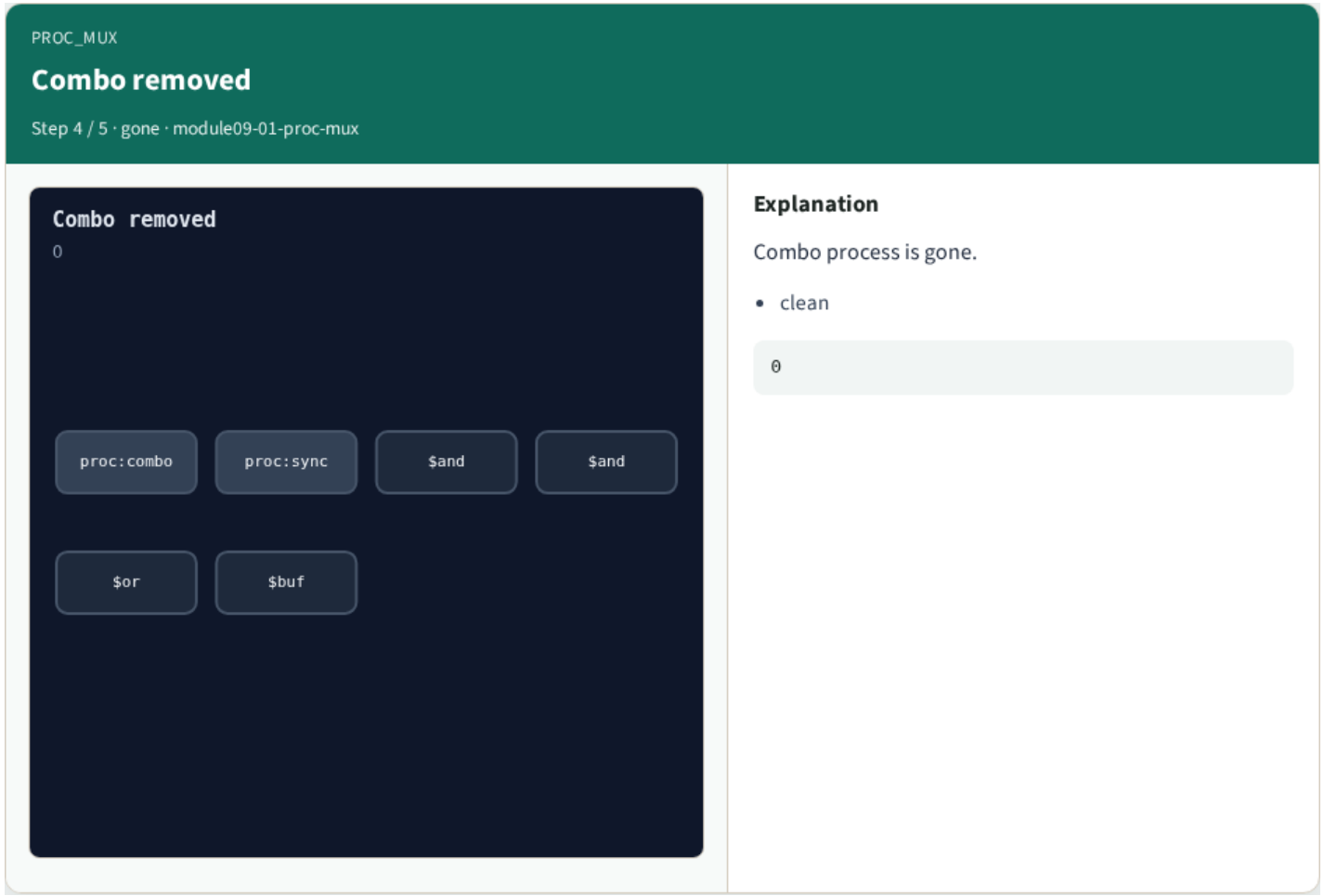
Emit $mux

S/A/B/Y


PROC_MUX
S/A/B/Y
Step 3 / 5 · ports · module09-01-proc-mux
S/A/B/Y
s
proc:combo
proc:sync
$and
$and
$or
$buf
Explanation
Sel s; A=a; B=b; Y=y.
ports
s

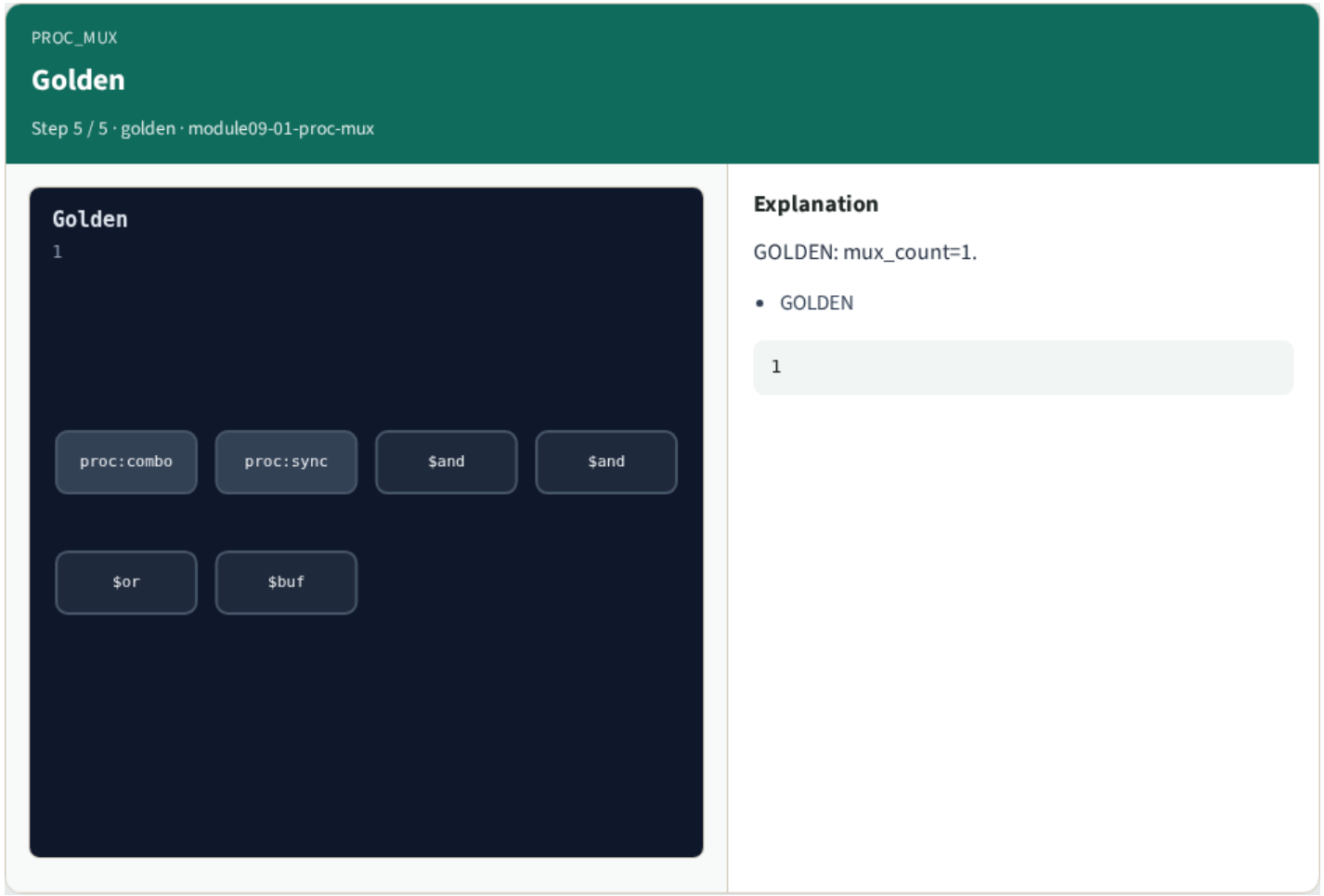
S/A/B/Y

Combo removed



Combo removed

Golden



Golden

Browser lab track

- Open **proc-mux**
- Replace the combo process with one mux cell and Check

Implement track

- Call algo_proc_mux(); assert mux_count == 1 and the cell type is dollar mux

Pitfalls

- Leaving the combo process in place
- Mapping sync processes as muxes

Try these

- From the common folder, run the algo solvers self-test, or call `algo_proc_mux` directly

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
for each combo process with two cases:
  emit dollar mux (S,A,B,Y)
  drop that process
ok if mux_count == 1
```

Next

- Complete the quiz for this part
- Next: proc_dff with async reset